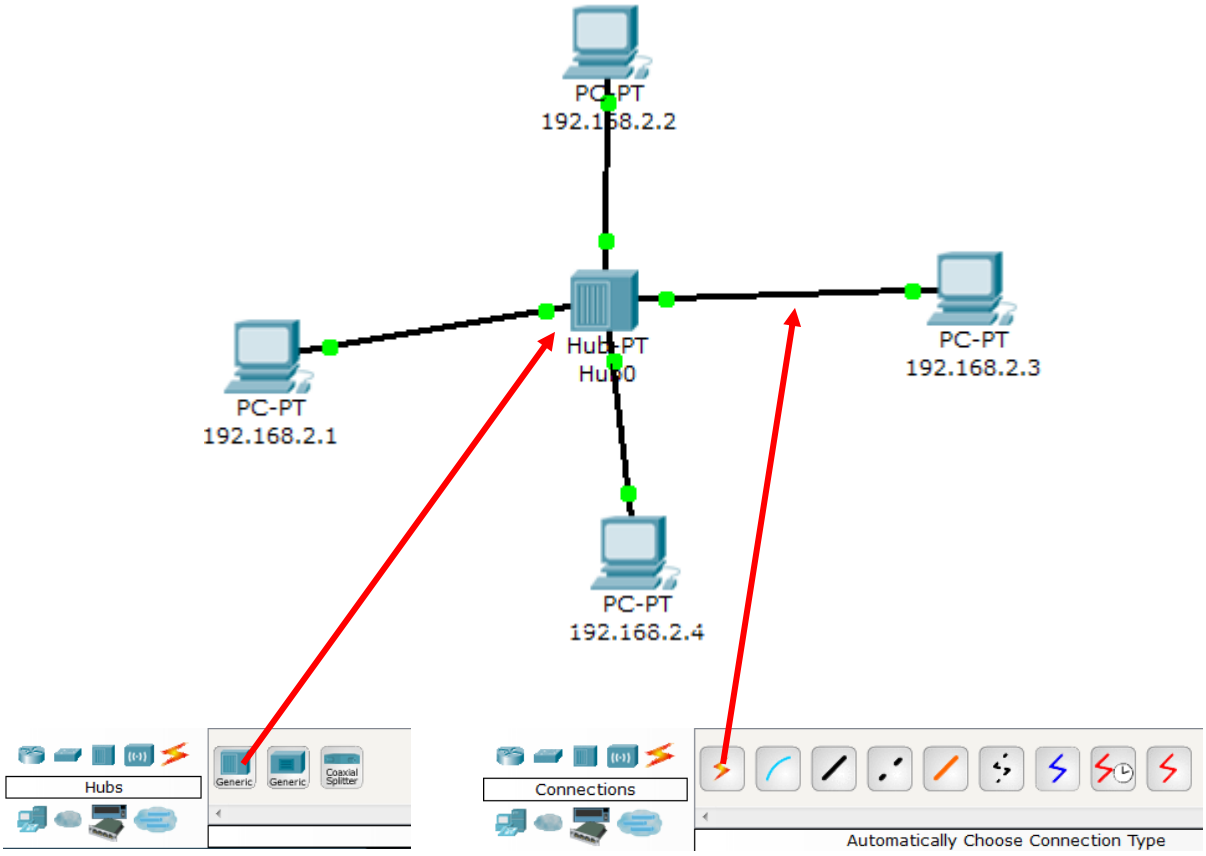
Lab 2: Introdtcion to Packet Tracer

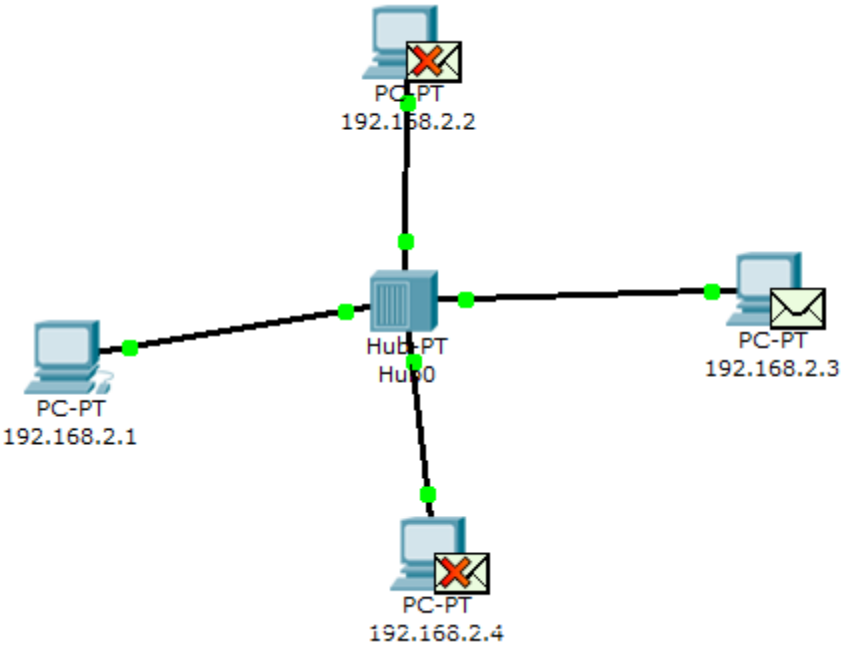
Basic of Packet Tracer:

Packet Tracer is widely used network simulator specially used to train students under CCNA **(Cisco Certified Network Associate)** and CCNP (**Cisco Certified Network Professional**) certification examination. Using this simulator student can get the real flavor to configure network components without buying real Cisco routers or switches.

Example-1

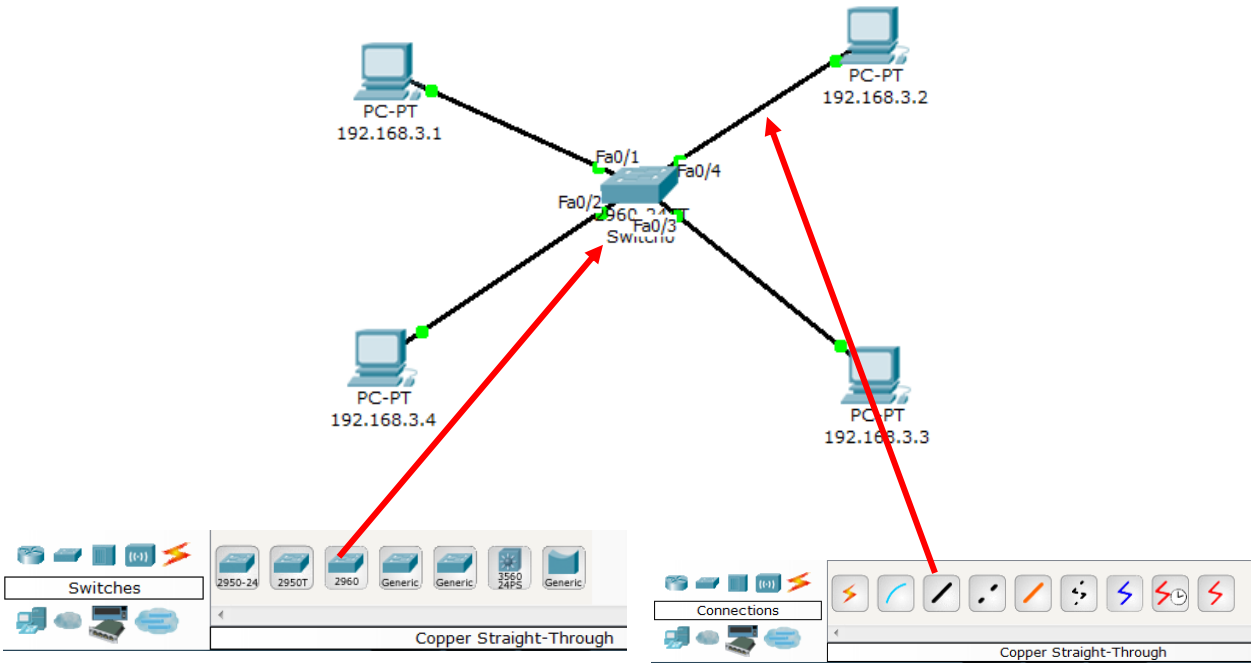
A HUB works a repeater and transmits a packet in all directions but only one receiver will accept it will be observed from the following network. Here *netid* of all the machine will be same i.e. they form a subnet. You can verify the network with ping command.

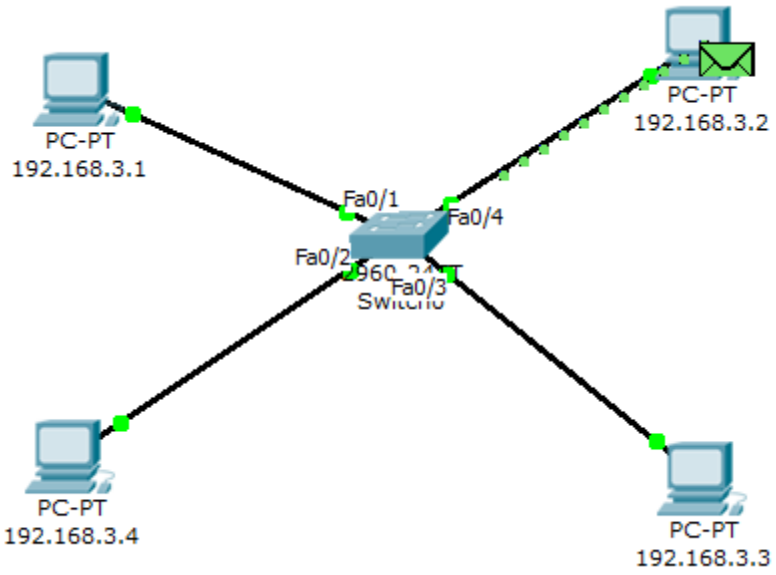




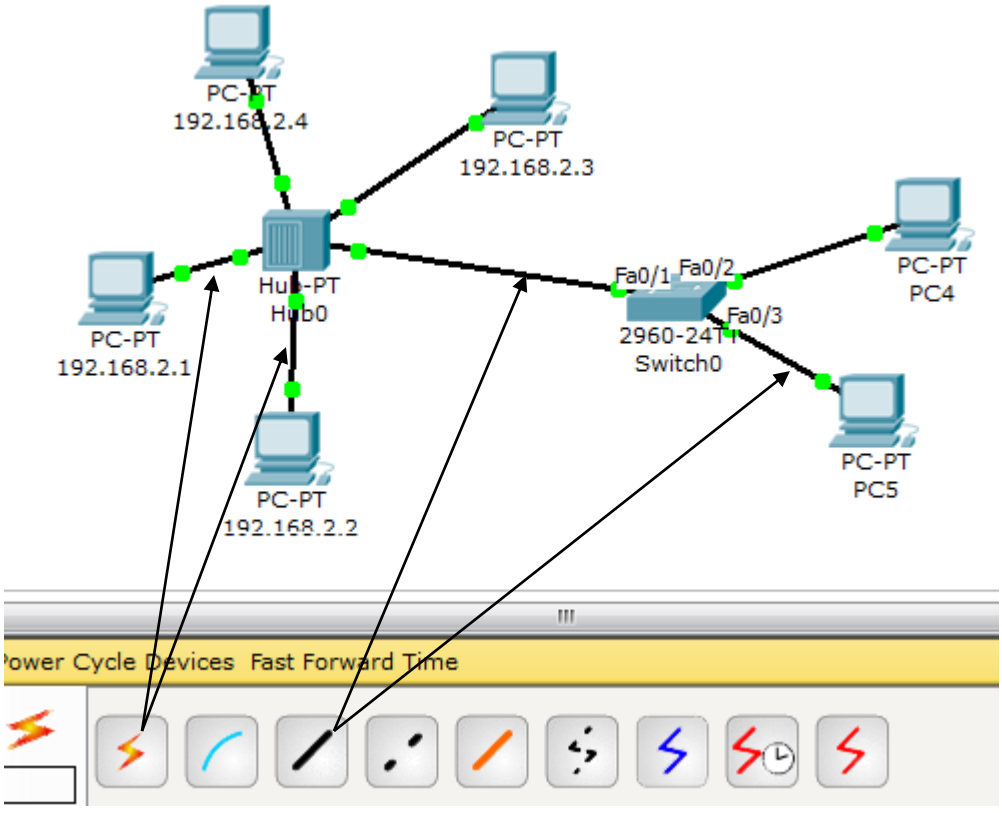
Example-2

A switch transmits a packet to only one receiver will be observed from the following network. Here *netid* of the entire machine will be same i.e. they form a subnet. You can verify the network with ping command. You have to wait for one minute to make the switch interface on.



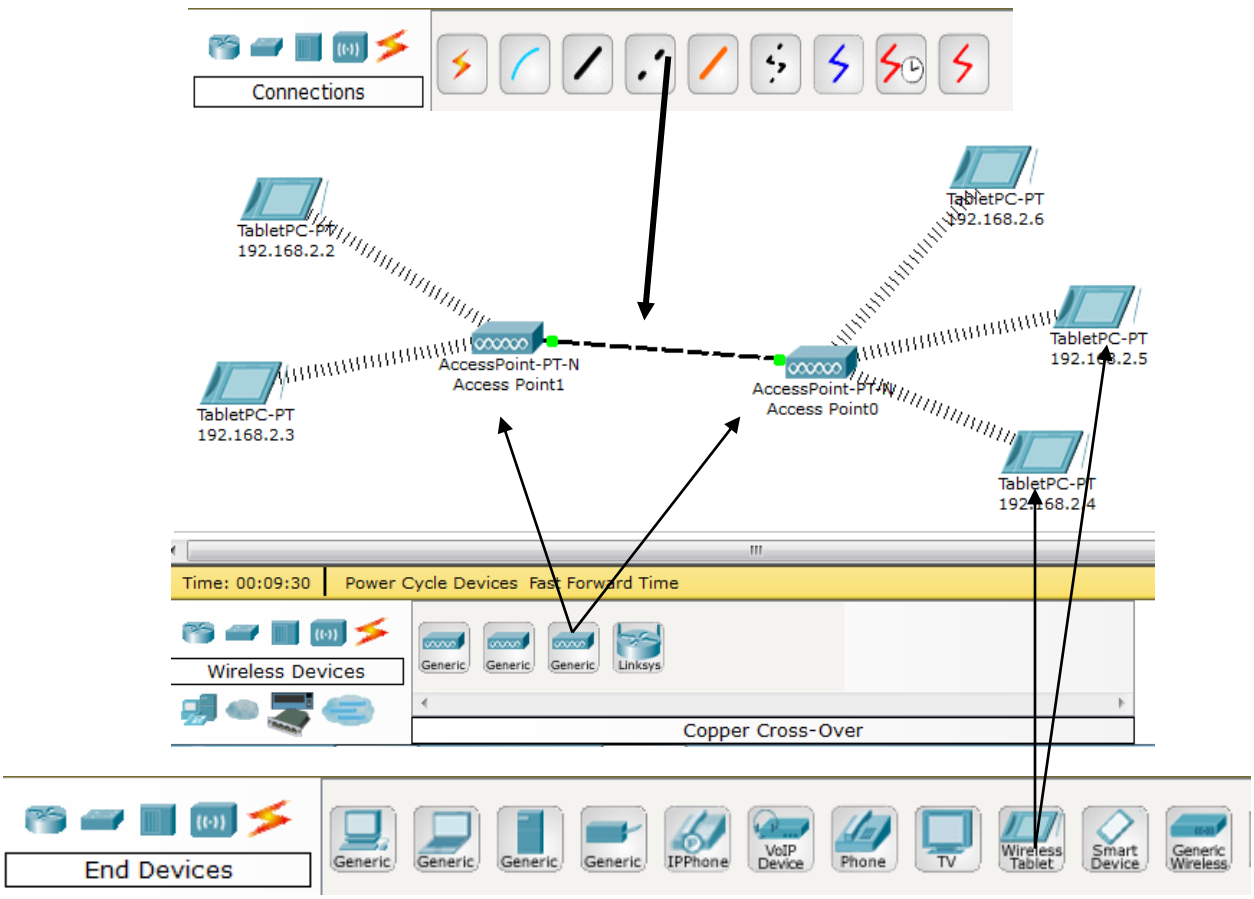


Example-3



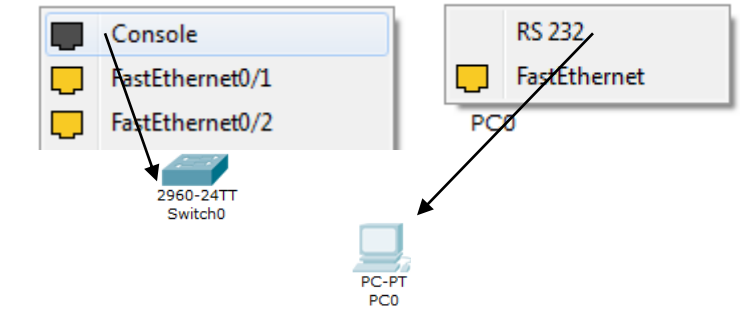
Example-4

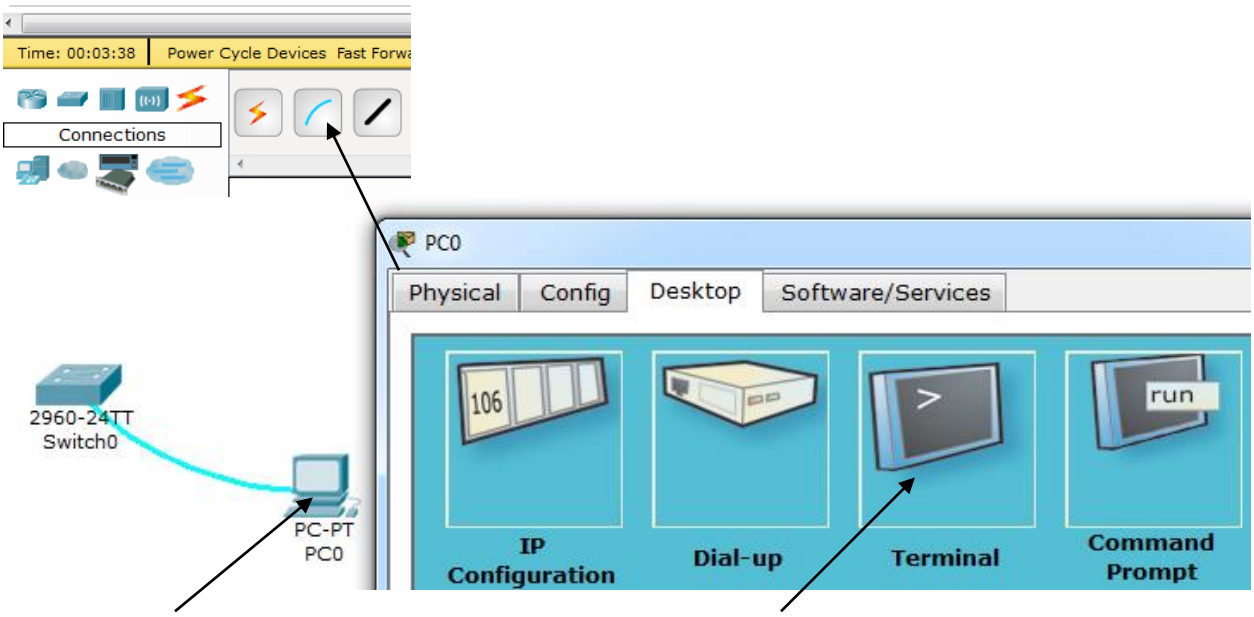
Wireless Network: Verify the with both ping and simulation.

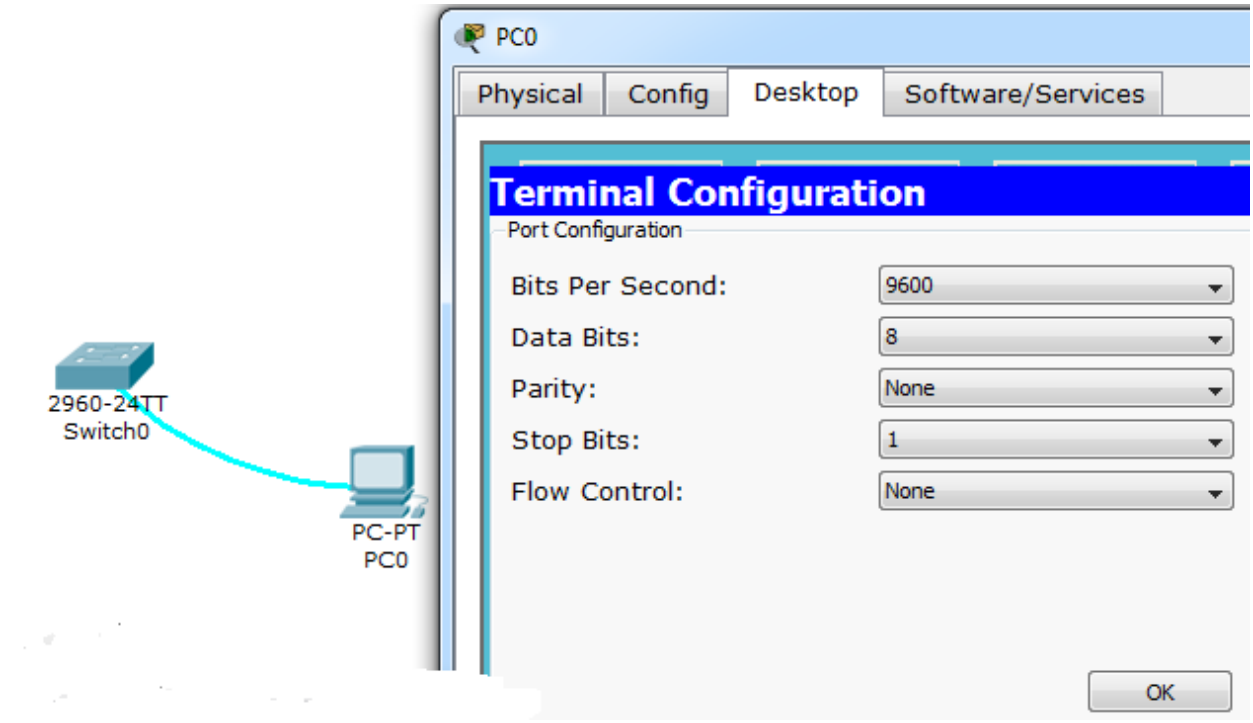

Connections
TabletPC-PT
192.168.2.2
TabletPC-PT
192.168.2.6
TabletPC-PT
192.168.2.5
AccessPoint-PT-N
Access Point1
AccessPoint-PT-N
Access Point0
TabletPC-PT
192.168.2.3
TabletPC-PT
192.168.2.4
Time: 00:09:30
Power Cycle Devices
Fast Forward Time
Wireless Devices
Generic
Generic
Generic
Linksys
Copper Cross-Over
End Devices
Generic
Generic
Generic
Generic
IPPhone
VoIP Device
Phone
TV
Wireless Tablet
Smart Device
Generic Wireless

Example-3

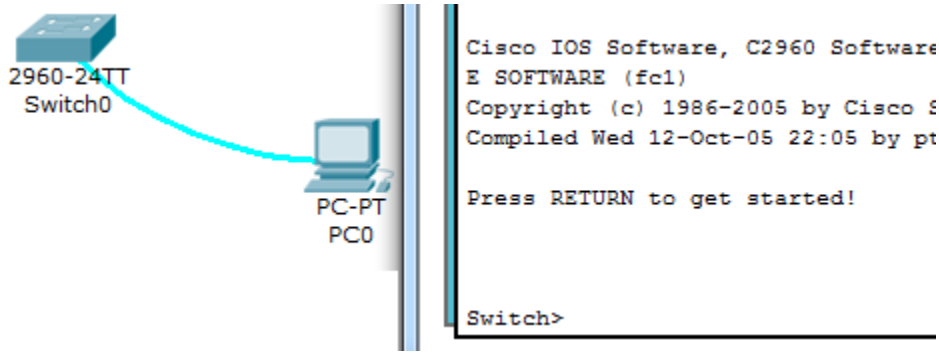
How to configure a switch using Laptop/PC and console port of a switch?







Now you can configure the switch on the command prompt of PC.



In a real network you can do the job using accessories/communication/hyperlink.

Let us put device enable password as **ju123** and console line password as **cseju**.

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#enable secret ju123
Switch(config)#line console 0
Switch(config-line)#password cseju
Switch(config-line)#login
Switch(config-line)#end
Switch# copy r startup-config
%SYS-5-CONFIG_I: Configured from console by console
Switch#exit
```

Password:

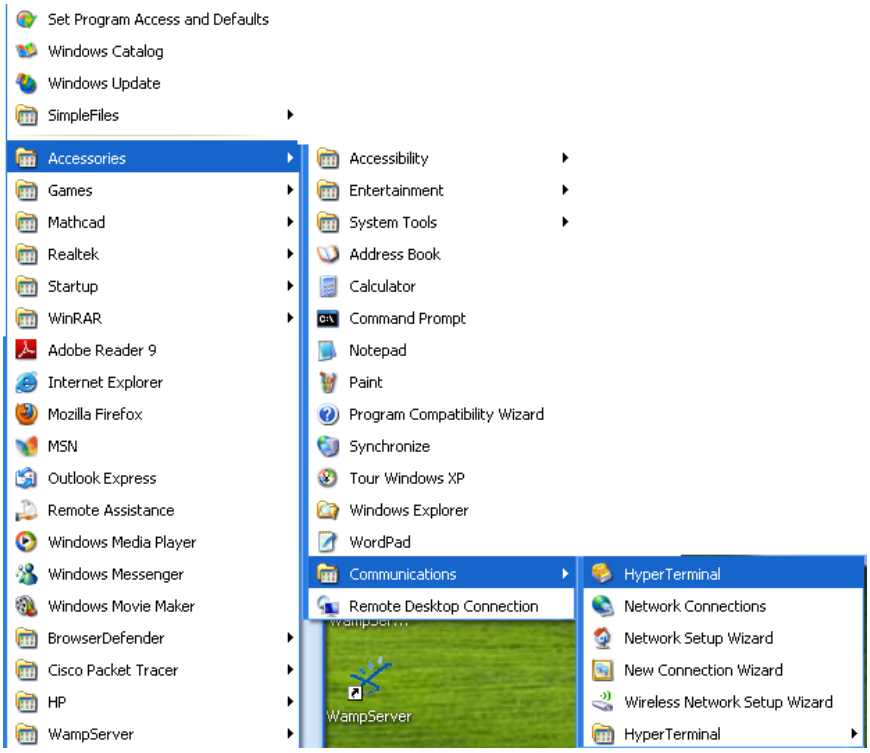
When you again get connected with switch through console cable and PC you have to provide both the passwords like:

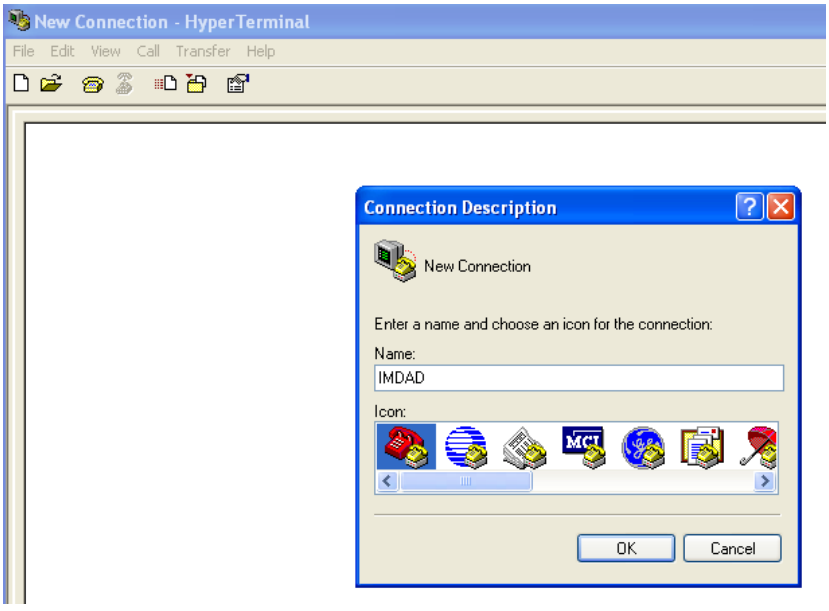
Password: cseju %console password
Switch>en
Password: ju123 %enable password
Switch#

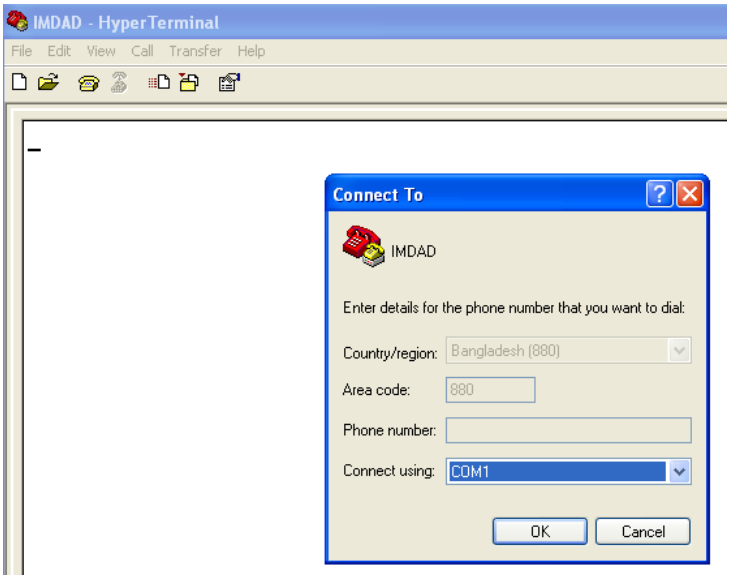
Let us now provide Telnet password to the device.

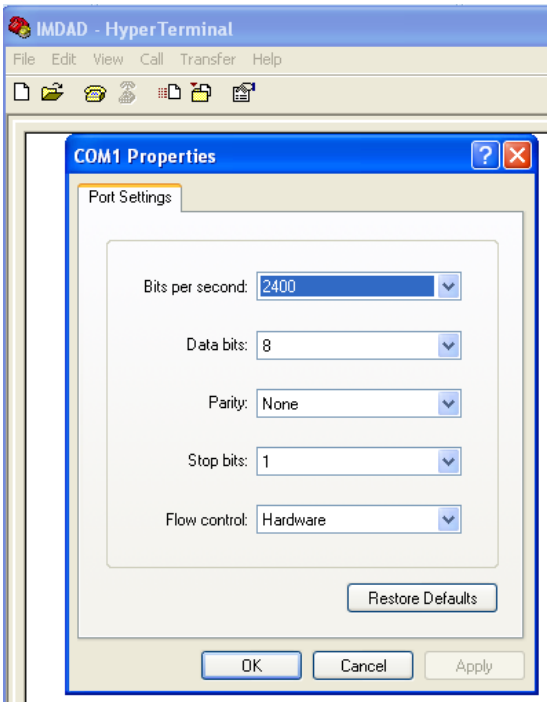
Switch>en
Password: ju123 %provide enable password
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#line vty 0 4 % 0 to 4 i.e. 5 users can access the device simultaneously
Switch(config-line)#password ***imdad***
Switch(config-line)#end
Switch#copy run startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Switch#

How to connect a PC/Laptop with console terminal of a switch/router in real life?

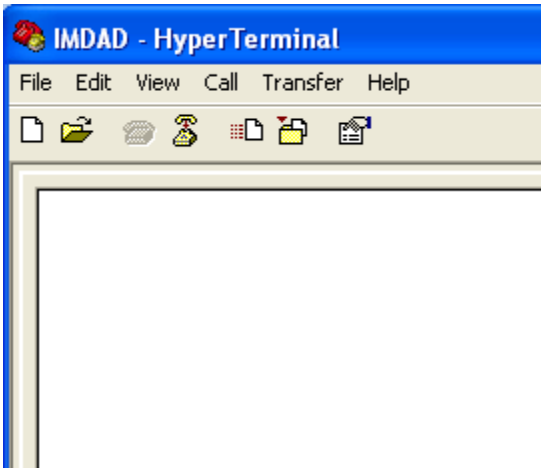

Set Program Access and Defaults
Windows Catalog
Windows Update
SimpleFiles
Accessories
Games
Mathcad
Realtek
Startup
WinRAR
Adobe Reader 9
Internet Explorer
Mozilla Firefox
MSN
Outlook Express
Remote Assistance
Windows Media Player
Windows Messenger
Windows Movie Maker
BrowserDefender
Cisco Packet Tracer
HP
WampServer
Accessibility
Entertainment
System Tools
Address Book
Calculator
Command Prompt
Notepad
Paint
Program Compatibility Wizard
Synchronize
Tour Windows XP
Windows Explorer
WordPad
Communications
Remote Desktop Connection
HyperTerminal
Network Connections
Network Setup Wizard
New Connection Wizard
Wireless Network Setup Wizard
HyperTerminal
WampServer


New Connection - HyperTerminal
File Edit View Call Transfer Help
Connection Description
New Connection
Enter a name and choose an icon for the connection:
Name:
IMDAD
Icon:
OK
Cancel

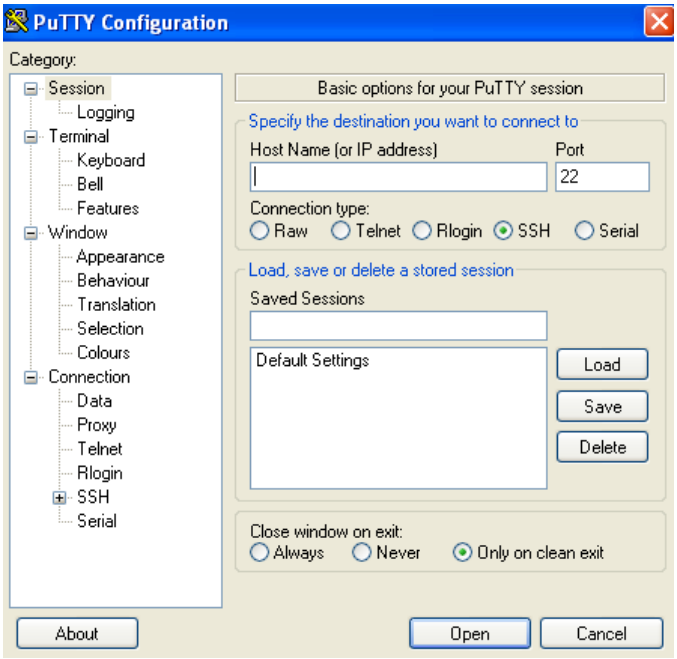




Now you will get the prompt for configuration.



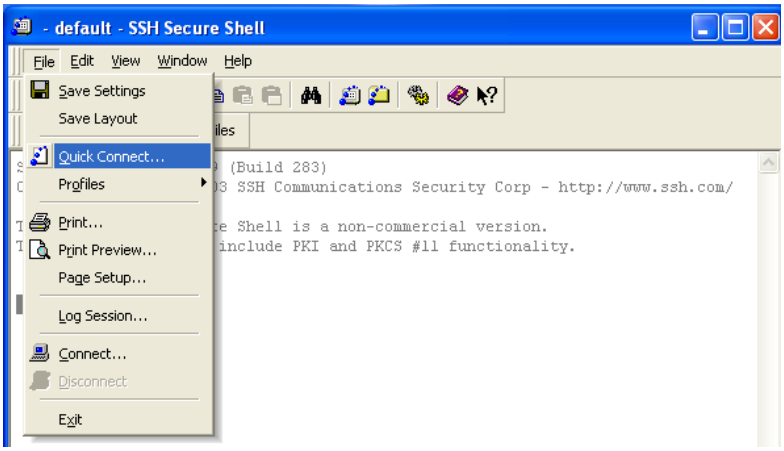
How to configure a switch/router from a remote PC/Laptop?
Use Putty or SSHSecureShellClient-3.2.9b283.

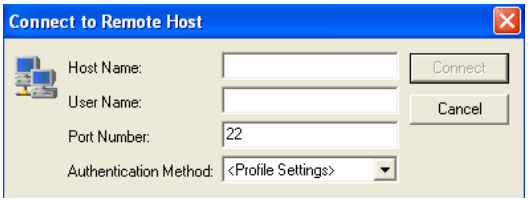


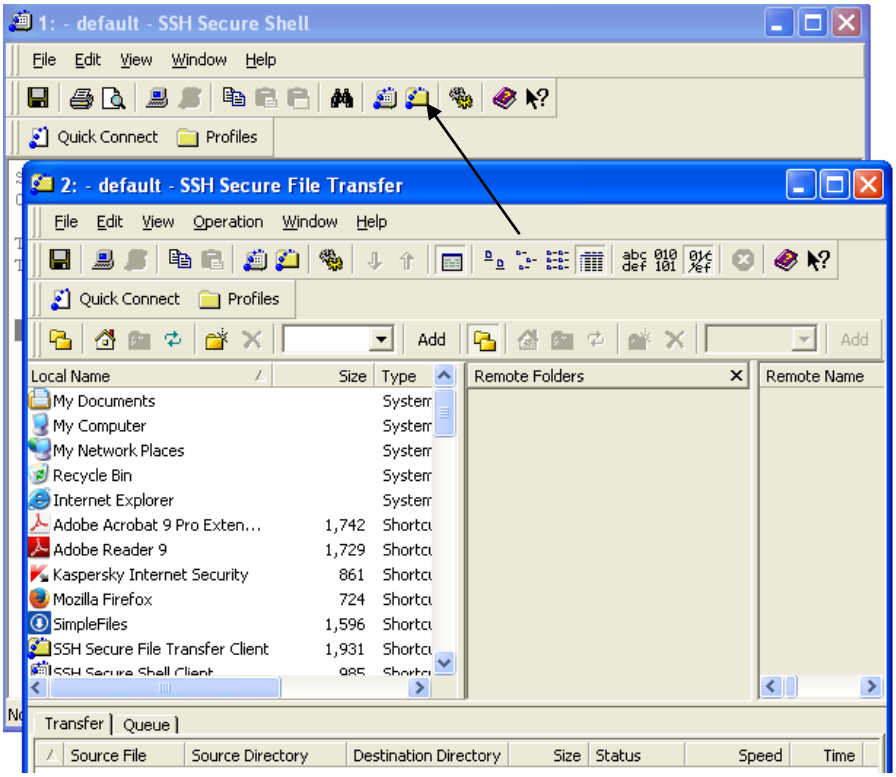
Putting IP address and click on open you will get the prompt of that switch/router.



Using SSH Secure Shell:







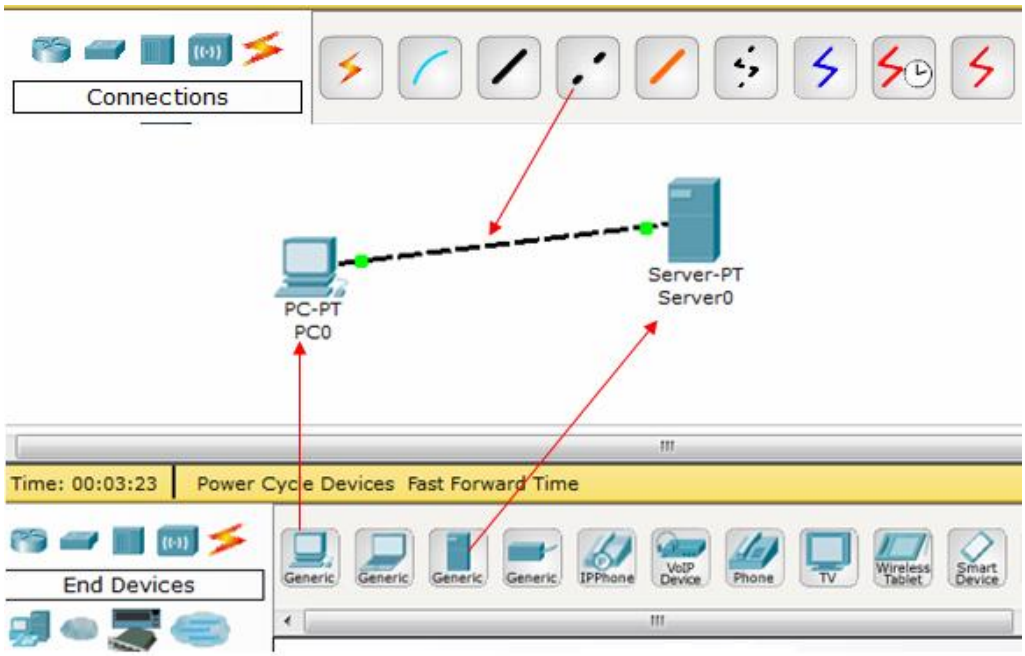
Now you can copy files or folder from remote machine.

Expt-1
Basic Client Server Model

Objective of this experiment is to verify the ICMP packet delivery between client-server networks. The server will be configured as a DNS server with a DNS name then a web page will be open with corresponding URL name. Finally *ping*, *tracert*, *ipconfig* and *arp-a* commands will be verified on the command prompt of both client and server.

Step-1

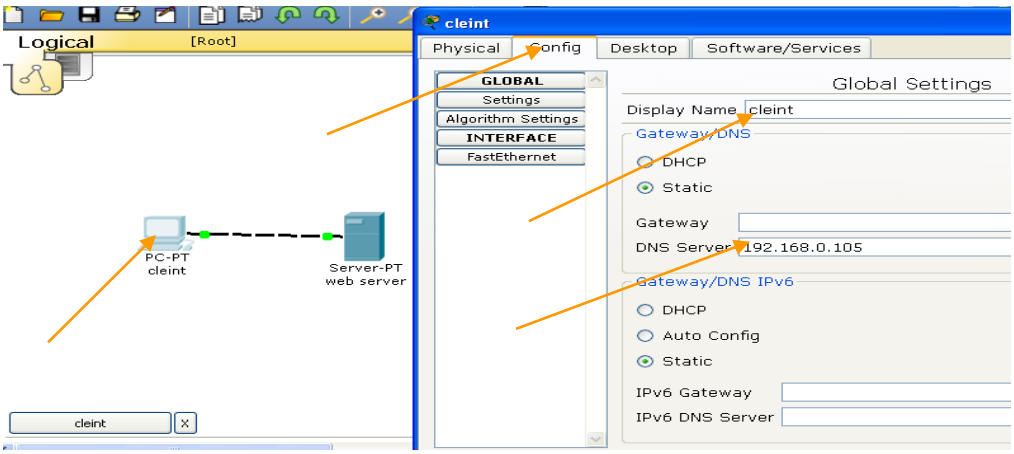
Implement the client-server network using the icon of client, server and connection (shown by arror) like below.



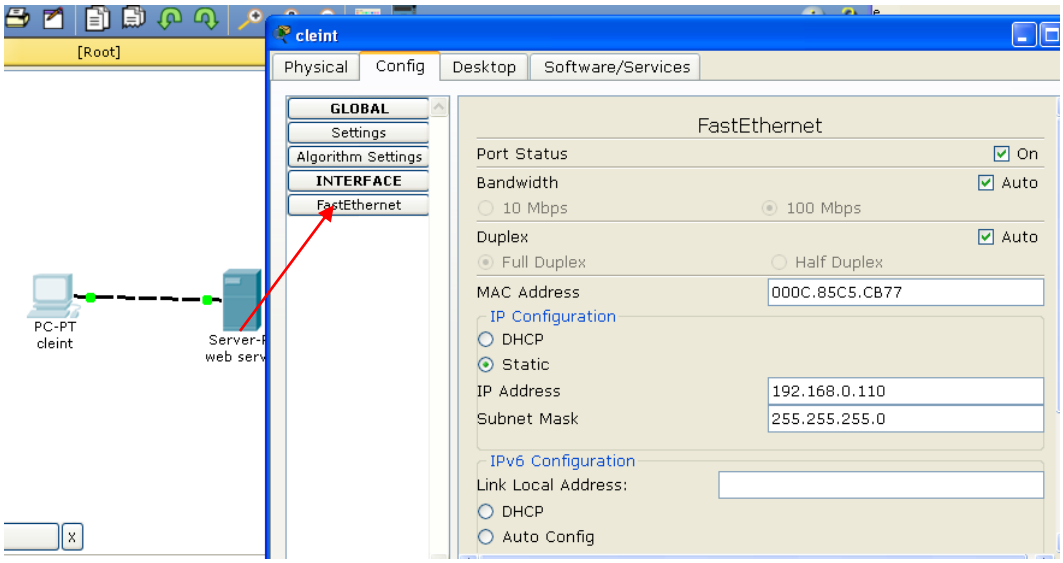
Step-2

Click on client and configure it using IP address of DNS server like below.

Step-3

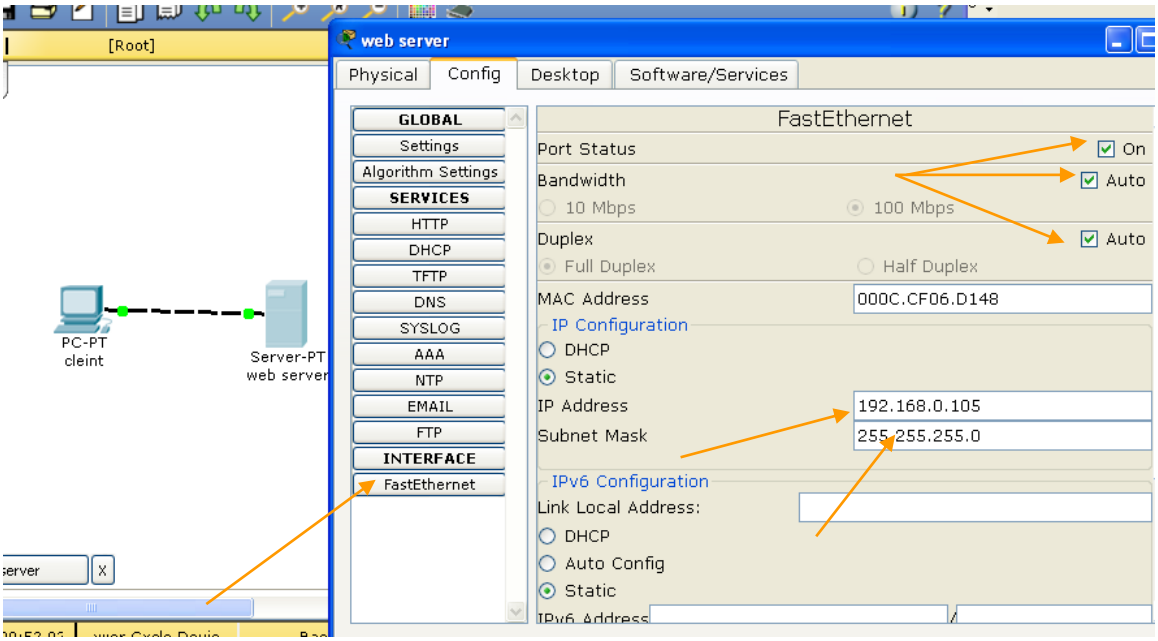


Click on FastEthernet and put IP address of the client like:



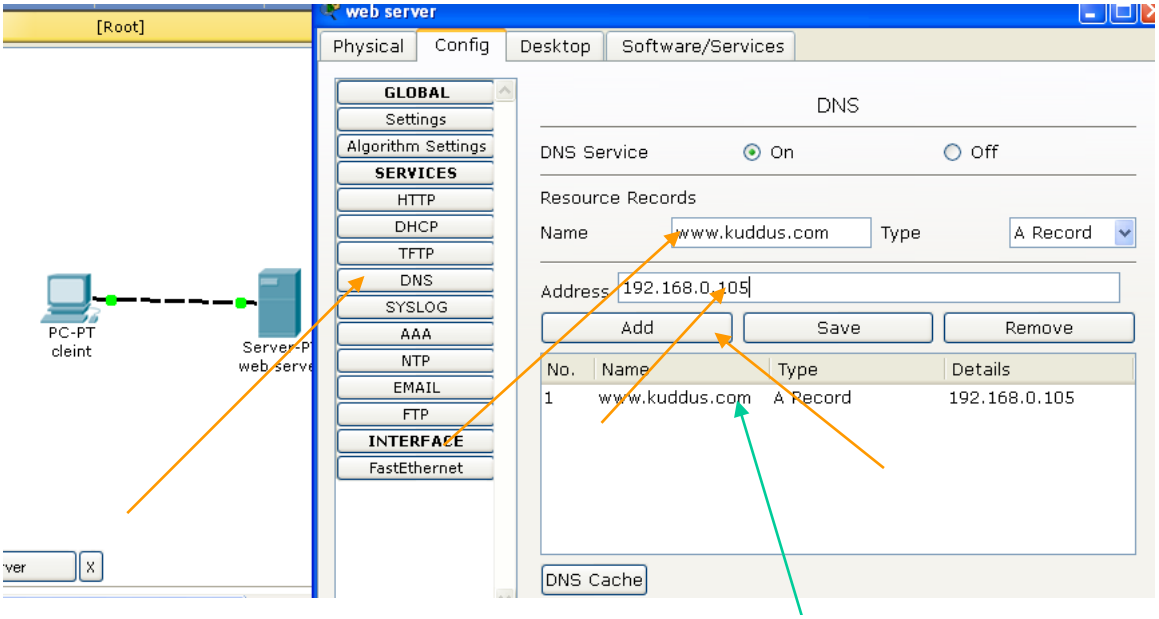
Step-4

Click on server, Config and FastEthernet then put IP address of the server like below.



Step-5

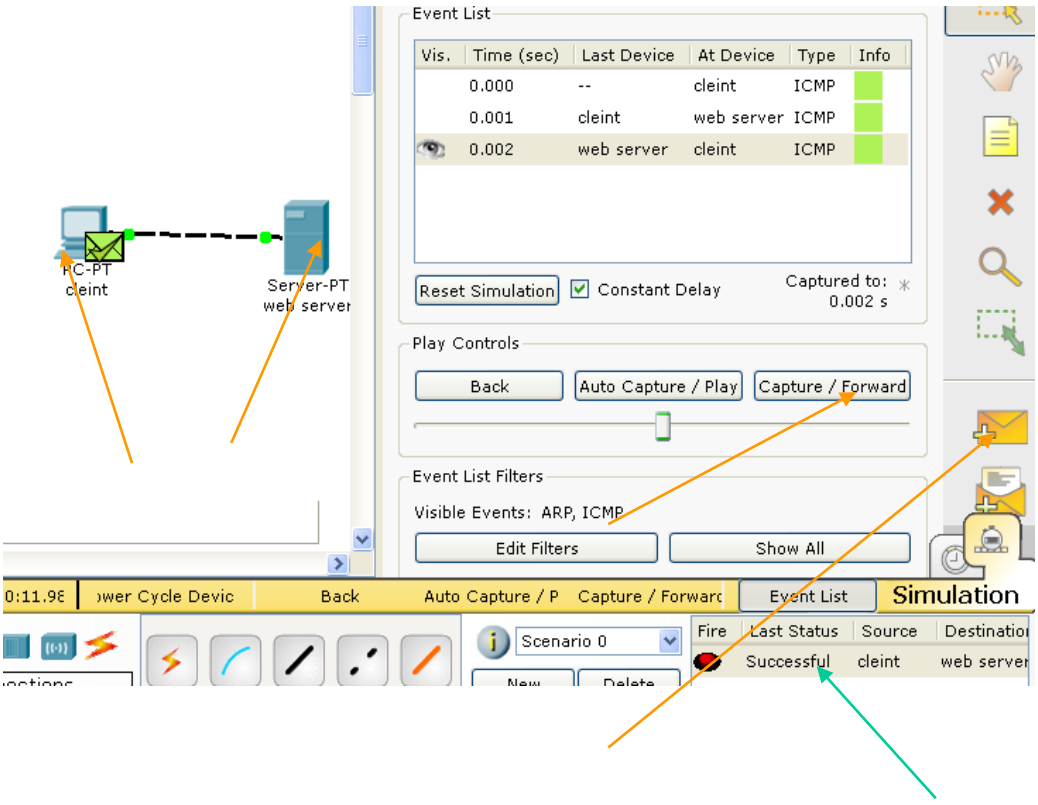
Click on DNS then put DNS name, IP address then click on add button.



You will get

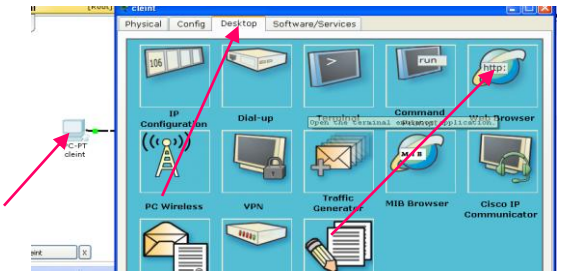
Step-6

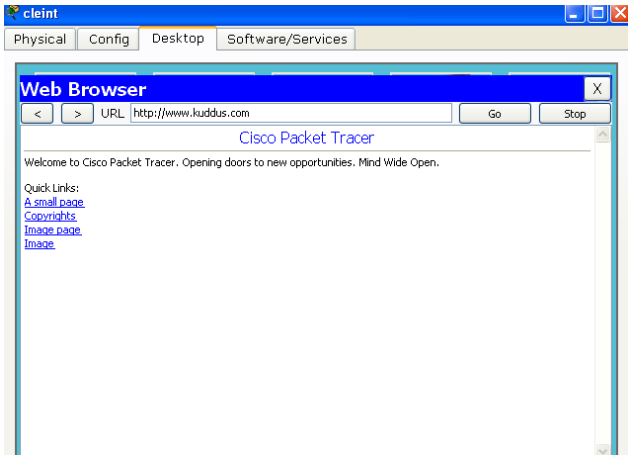
Verify the network by sending ICMP test packet under simulation mode.



Step-7

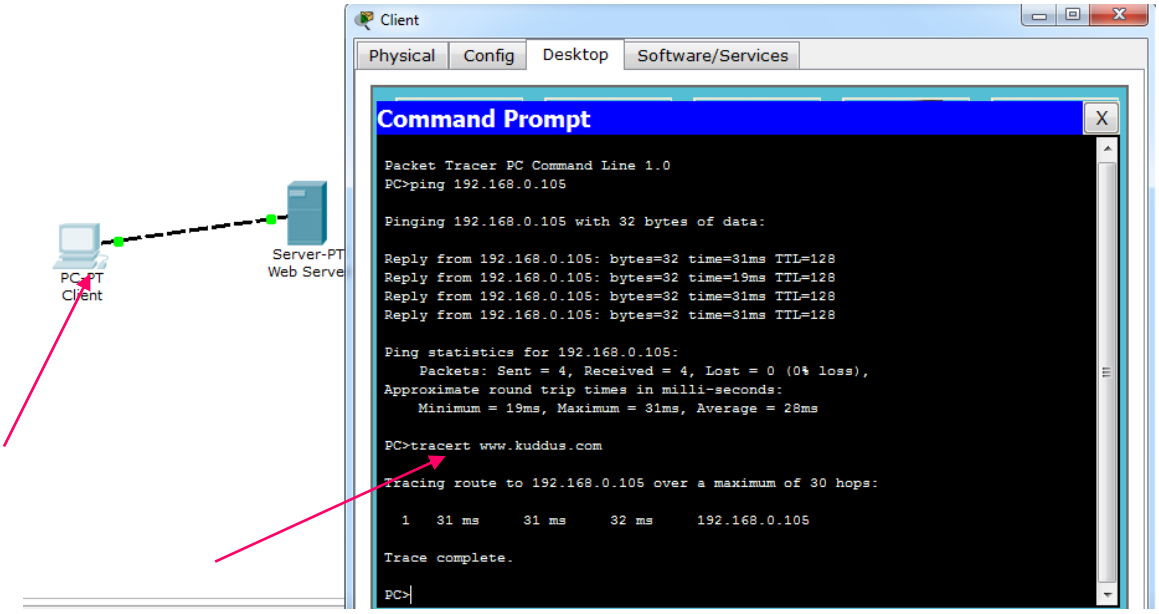
Click on client, desktop and web browser then write the URL of web page and click on Go. Verify whether you get the following result.





Step-8

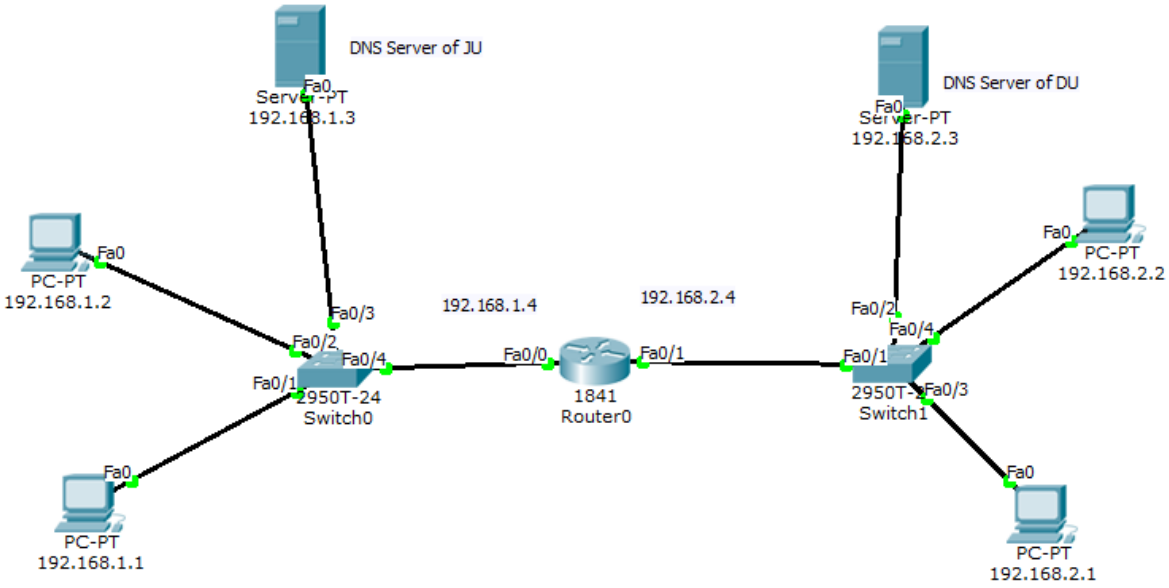
Verify *ping*, *tracert*, *ipconfig* and *arp-a* commands on the command prompt of the client.

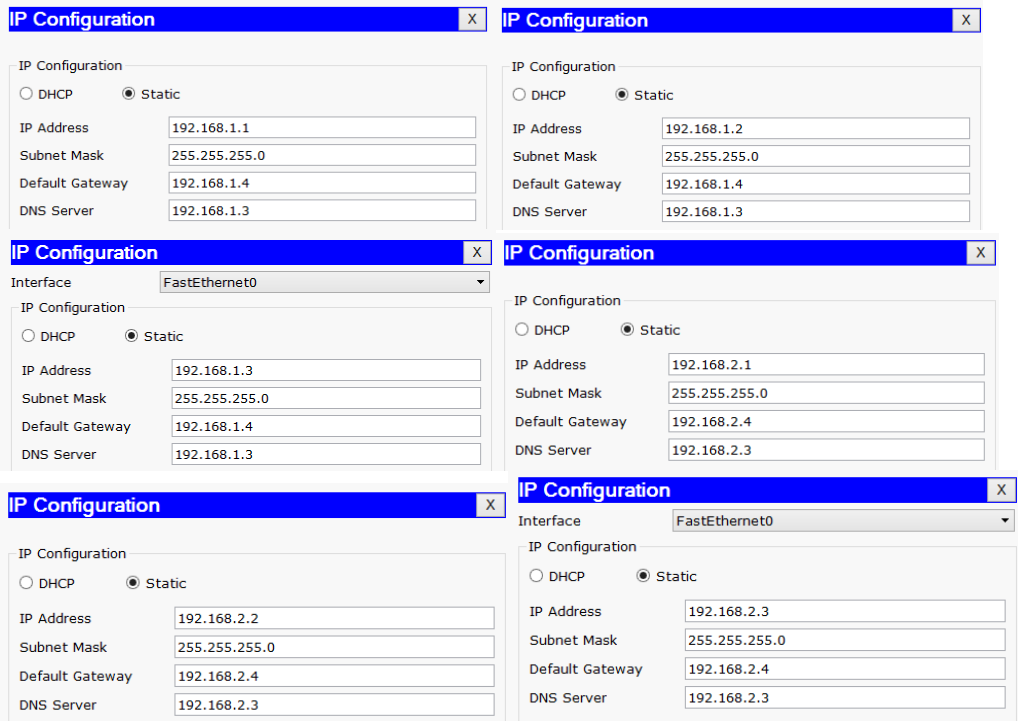


DNS Server Configuration

Step-1

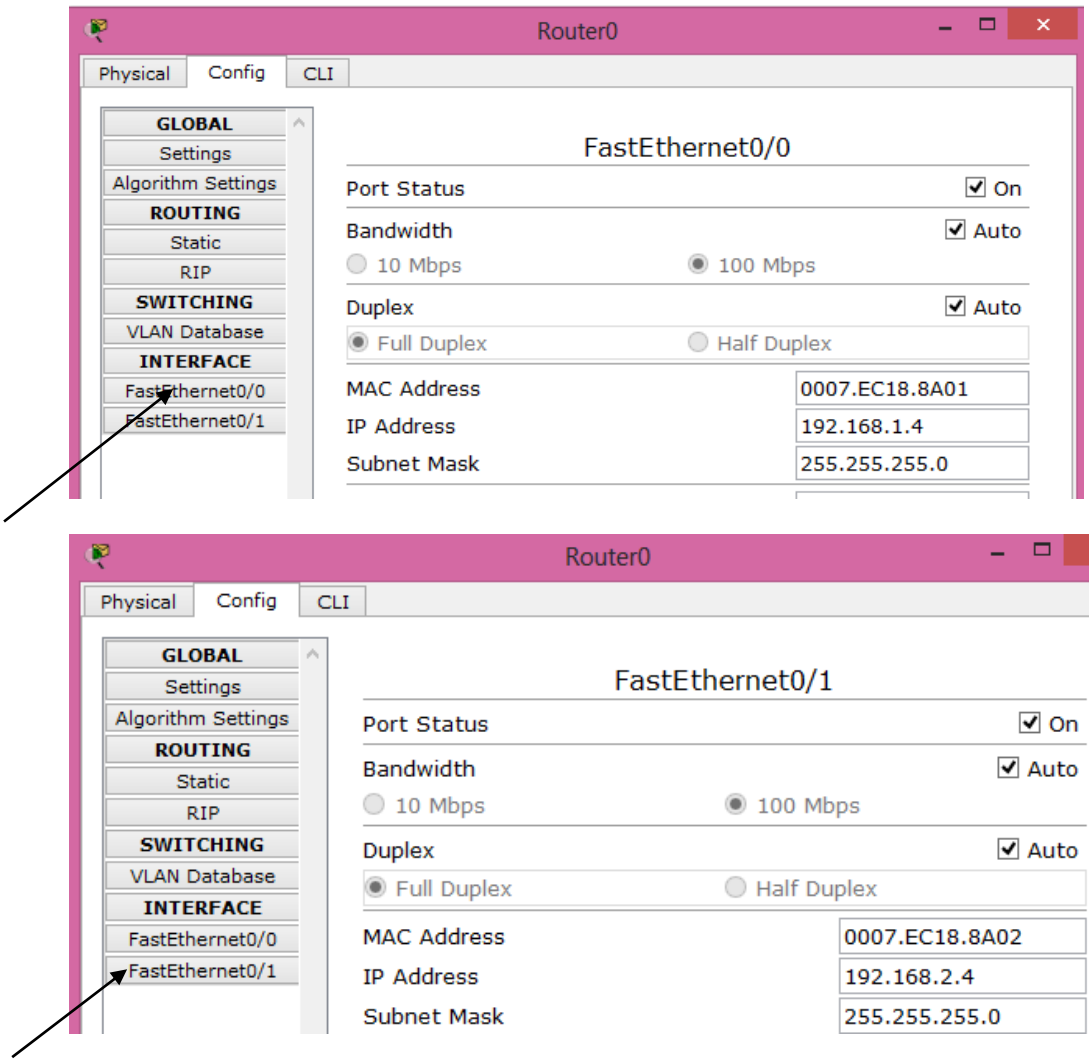
Implement the following network in Packet Tracer; where the IP, subnet mask, gateway and DNS server is shown below.



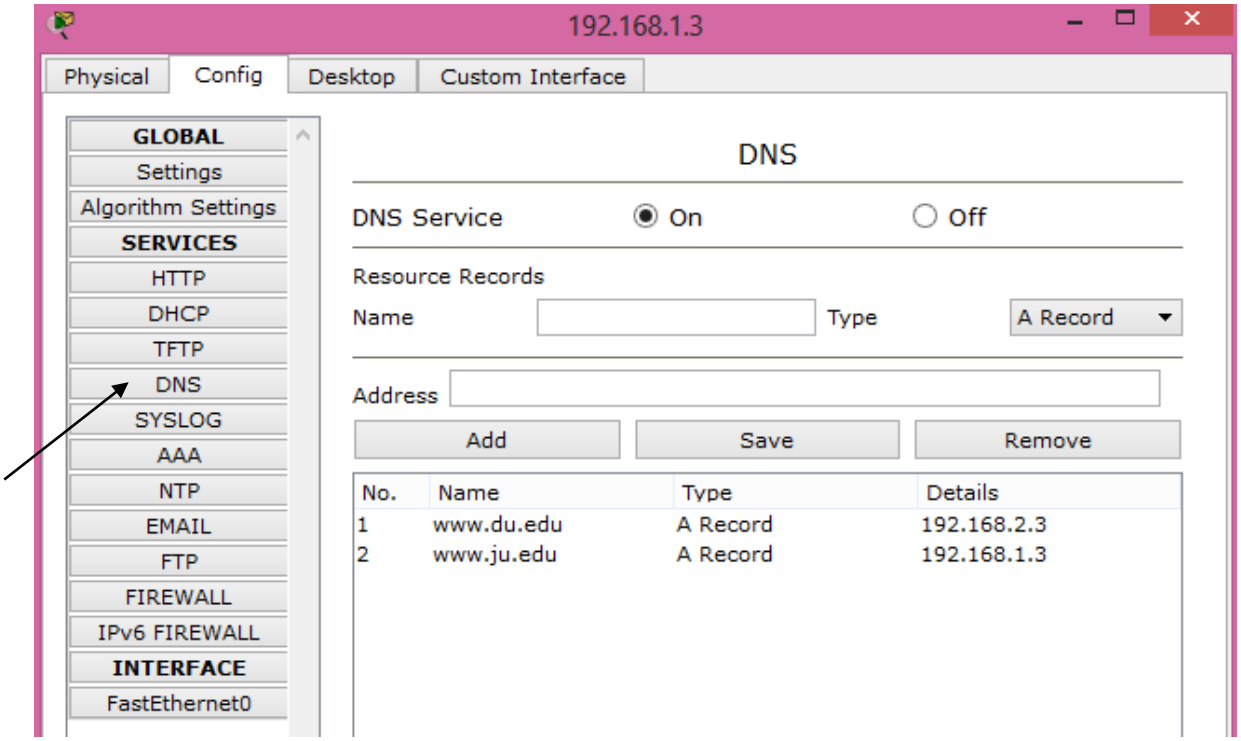


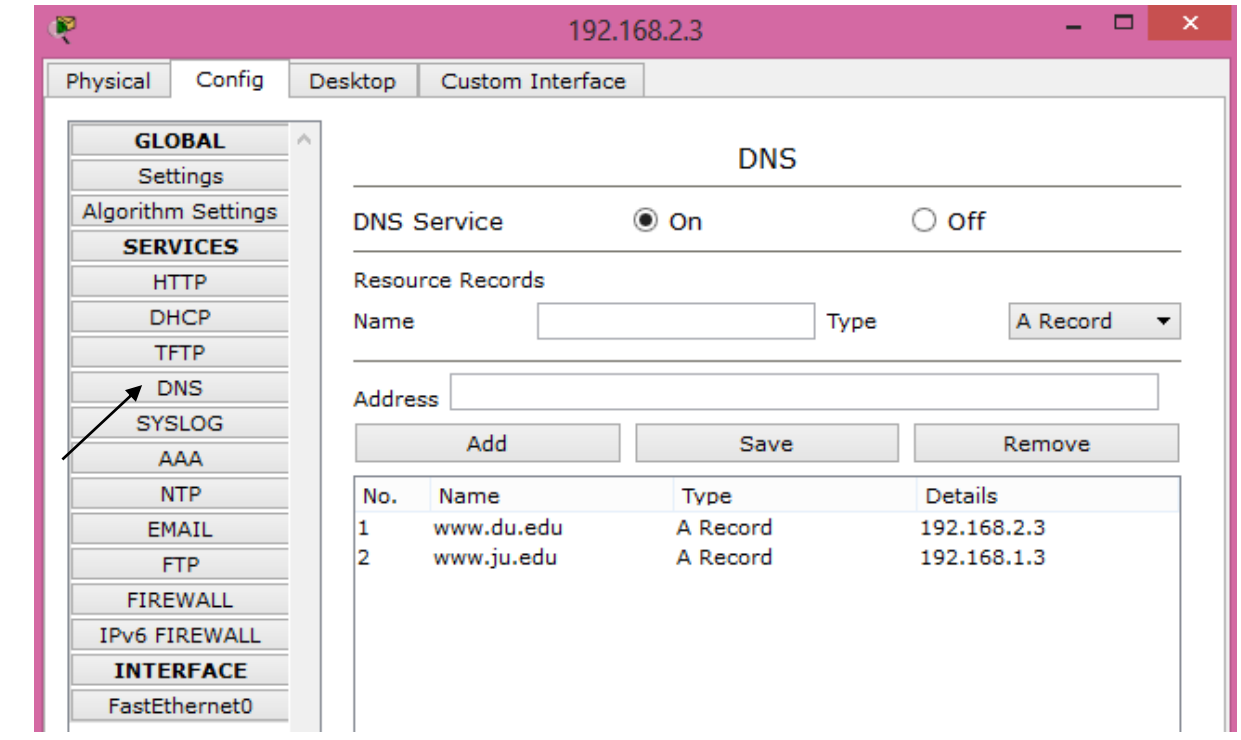
Step-2

Click on the router and put the appropriate IP address on both Fa 0/0 and Fa 0/1 and make them on



Step-3
Put DNS name and corresponding IP address on both the servers.


192.168.1.3
Physical
Config
Desktop
Custom Interface
GLOBAL
Settings
Algorithm Settings
SERVICES
HTTP
DHCP
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
FIREWALL
IPv6 FIREWALL
INTERFACE
FastEthernet0
DNS
DNS Service
On
Off
Resource Records
Name
Type
A Record
Address
Add
Save
Remove
No.
Name
Type
Details
1
www.du.edu
A Record
192.168.2.3
2
www.ju.edu
A Record
192.168.1.3

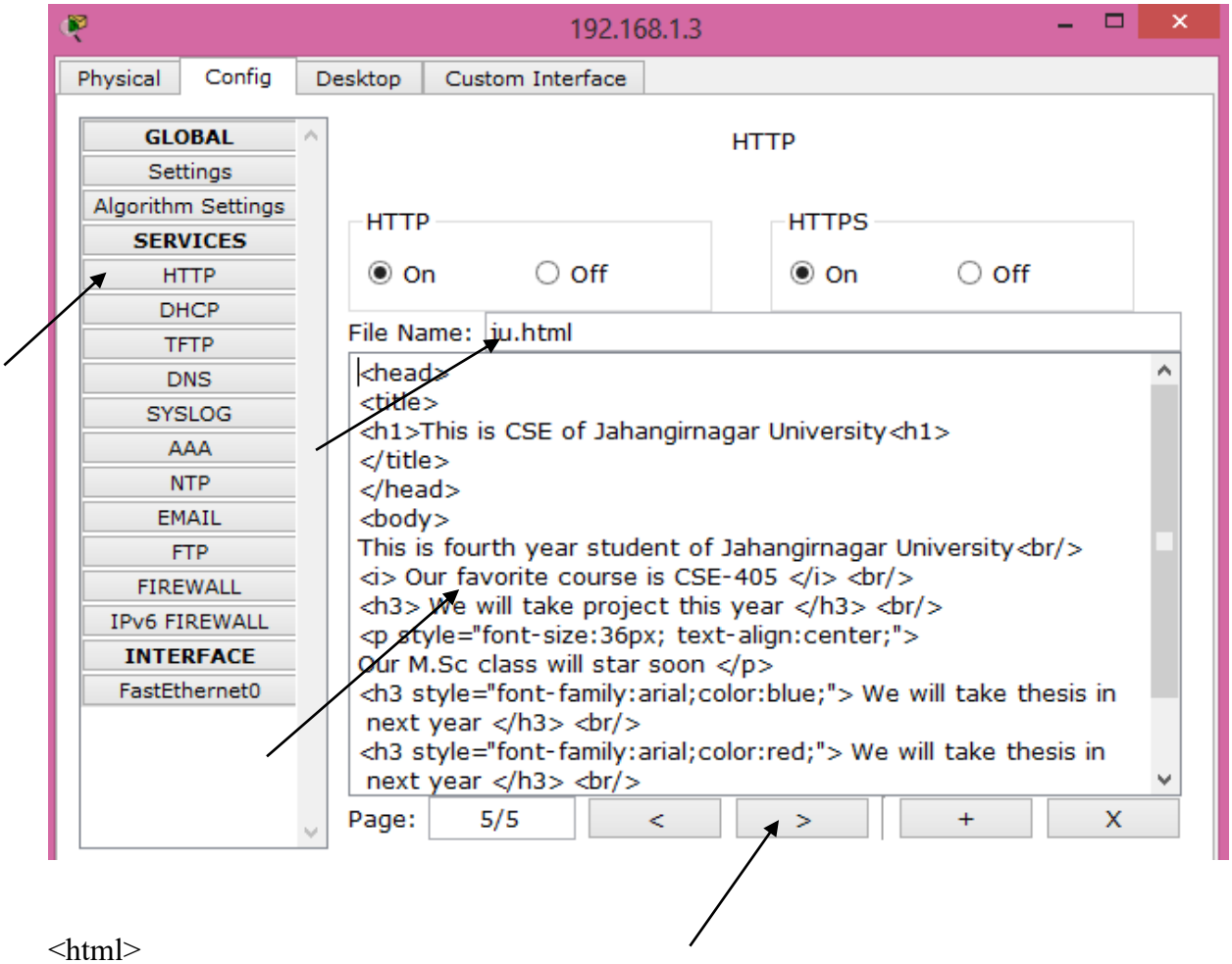

192.168.2.3
Physical
Config
Desktop
Custom Interface
GLOBAL
Settings
Algorithm Settings
SERVICES
HTTP
DHCP
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
FIREWALL
IPv6 FIREWALL
INTERFACE
FastEthernet0
DNS
DNS Service
On
Off
Resource Records
Name
Type
A Record
Address
Add
Save
Remove
No.
Name
Type
Details
1
www.du.edu
A Record
192.168.2.3
2
www.ju.edu
A Record
192.168.1.3

<html>
<head>

<title>
<h1>This is CSE of Jahangirnagar University<h1>
</title>
</head>
<body>
This is fourth year student of Jahangirnagar University
 Our favorite course is CSE-405
<h3> We will take project this year </h3>

Our M.Sc class will star soon
<h3 style="font-family:arial;color:blue;"> We will take thesis in next year </h3>
<h3 style="font-family:arial;color:red;"> We will take thesis in next year </h3>
</body>
</html>

Load the above HTML code on JU DNS Server

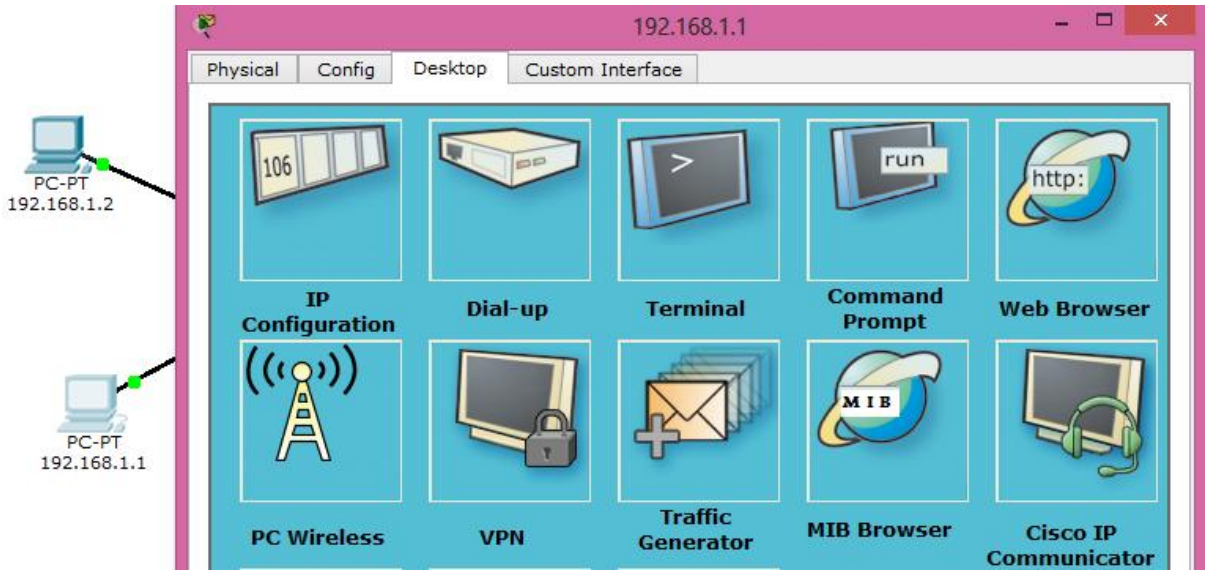


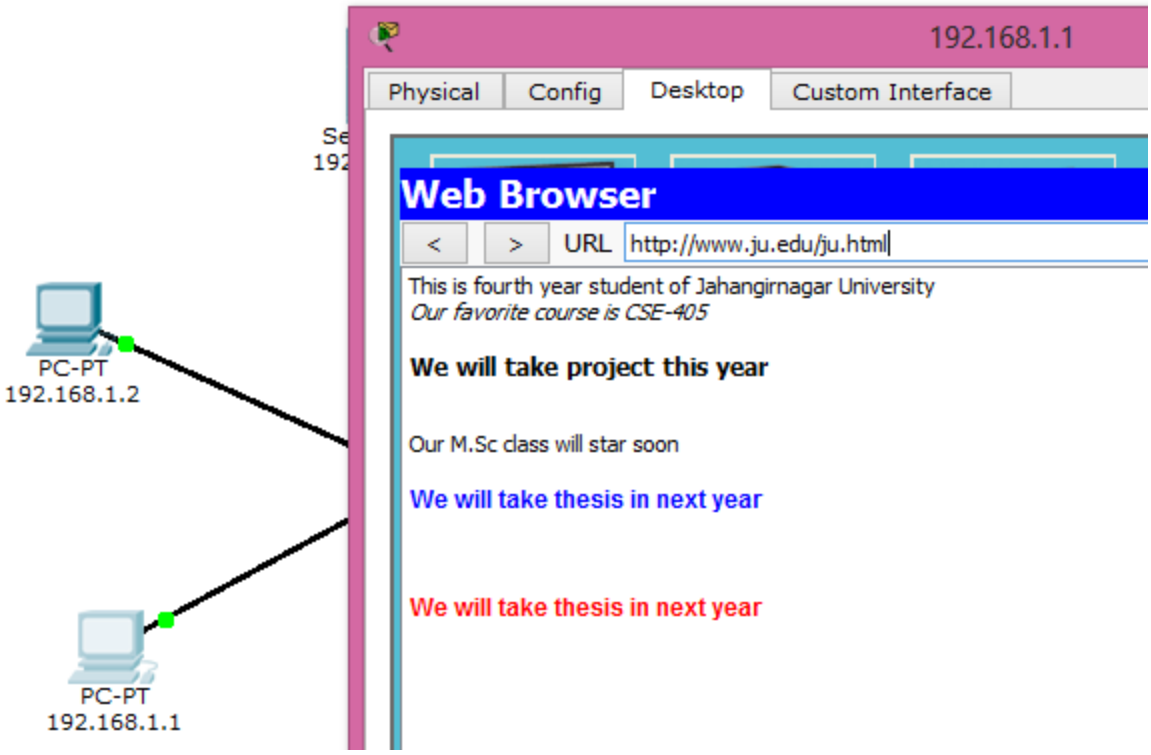
<html>
<head>
<title>
<h1>This is CSE of Dhaka University<h1>
</title>

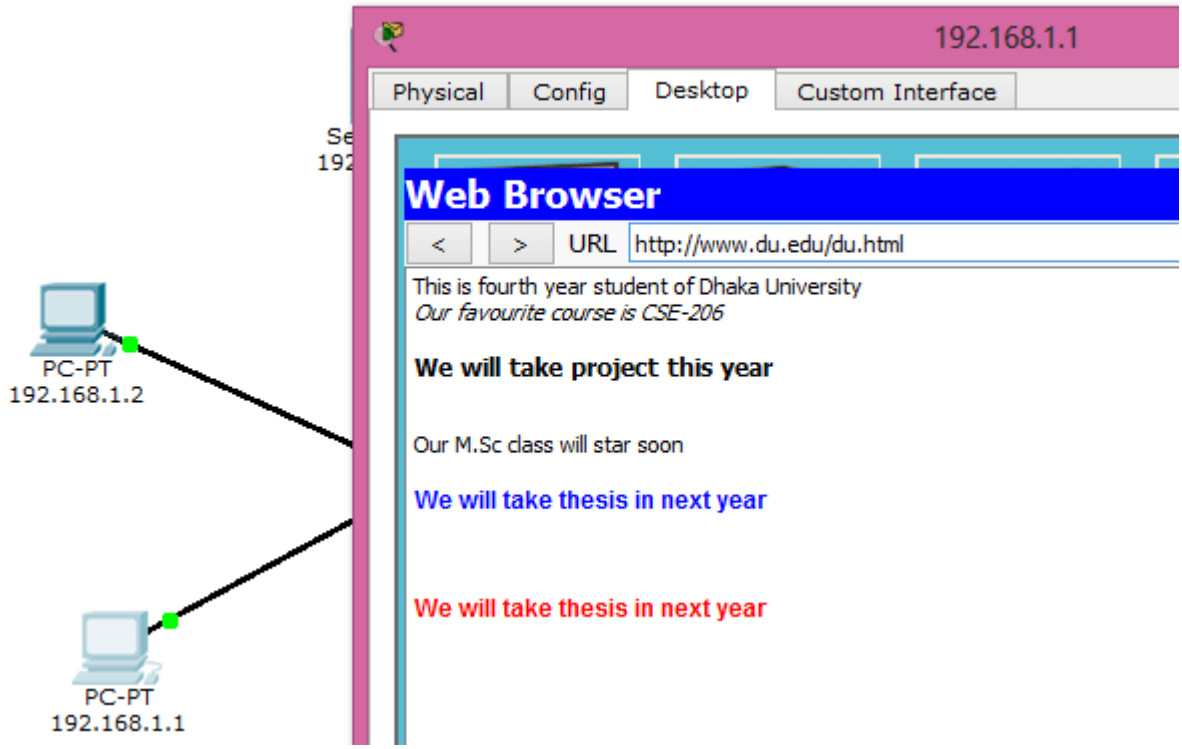
```
</head>
<body>
This is fourth year student of Dhaka University<br/>
<i> Our favourite course is CSE-206 </i> <br/>
<h3> We will take project this year </h3> <br/>
<p style="font-size:36px; text-align:center;">
Our M.Sc class will star soon </p>
<h3 style="font-family:arial;color:blue;"> We will take thesis in next year </h3> <br/>
<h3 style="font-family:arial;color:red;"> We will take thesis in next year </h3> <br/>
</body>
</html>
```

Load the above HTML code on DU DNS Server

Now Browse the URL page of each server from any terminal.




192.168.1.1
Physical
Config
Desktop
Custom Interface
Web Browser
URL
http://www.ju.edu/ju.html
This is fourth year student of Jahangirnagar University
Our favorite course is CSE-405
We will take project this year
Our M.Sc class will star soon
We will take thesis in next year
We will take thesis in next year
PC-PT
192.168.1.2
PC-PT
192.168.1.1


192.168.1.1
Physical
Config
Desktop
Custom Interface
Web Browser
URL
http://www.du.edu/du.html
This is fourth year student of Dhaka University
Our favourite course is CSE-206
We will take project this year
Our M.Sc class will star soon
We will take thesis in next year
We will take thesis in next year
PC-PT
192.168.1.2
PC-PT
192.168.1.1